

HOTEL BOOKING CANCELLATION **ANALYTICS AND RISK PREDICTION**

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Section 1: Hotel Bookings Analysis

Overview

Section 1 has three data quality checks on the hotel bookings dataset. Each check produces a specific number or short finding.

The dataset was loaded with `keep_default_na=False` to prevent the loyalty tier value 'None' from being read as a missing value. Without this, roughly 46% of loyalty tier data disappears silently. Date columns were converted to datetime, and `review_rating` was converted to numeric.

Four data quality flags were created before any filtering:

- Invalid stays: `checkout_date` on or before `checkin_date`
- Bookings before signup: `booking_date` earlier than `customer_signup_date`
- Zero-room bookings: `num_rooms` equal to zero
- Cancelled bookings with reviews: `booking_status` is Cancelled but `review_rating` is not null

Findings

A1. 120 bookings have invalid stay dates (checkout on or before check-in).

A2.

Customer Segment	Rating Range	Mean Rating
Corporate	3.0–10.0	7.23
Group	1.0–5.0	3.76
Individual	1.0–5.0	3.77

Implication: Corporate customers use a 1–10 rating scale, whereas Individual and Group customers use a 1–5 scale. As a result, raw average ratings are not directly comparable across customer segments and should be normalized before comparing satisfaction levels.

A3. Realized revenue for Luxury properties, after removing invalid stays, zero-room bookings, and booking-before-signup records, and keeping only Completed bookings, is **Rs 89,009,636.79**.

Cleaning Summary

After removing invalid stays, booking-before-signup records, and zero-room bookings, the dataset went from 12,000 rows to 11,661 rows (339 rows removed). Cancelled bookings with reviews had their `review_rating` and `review_date` cleared, but the booking row itself was kept. The loyalty tier string 'None' was preserved throughout as a valid category.

Cleaning Step	Action
Removed invalid stays	Rows excluded from clean dataset
Removed bookings before customer signup	Rows excluded from clean dataset

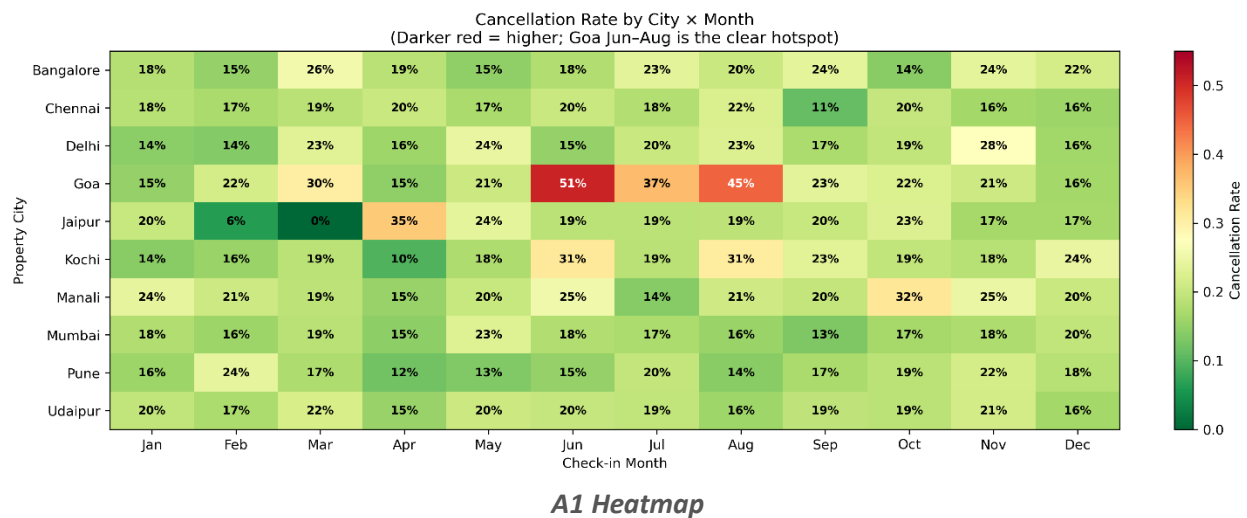
Removed zero-room bookings	Rows excluded from clean dataset
Cleared reviews on cancelled bookings	Review data cleared; booking row kept
Preserved loyalty tier 'None'	Treated as valid category, not null
Completed filter for realized revenue	Applied in A3 only; other statuses kept in clean dataset

Section 2: Cancellation Crisis Analysis

The platform's overall cancellation rate is around 19%, well above the 12% industry benchmark. The board wants to reduce this to 14% within two quarters. This section identifies where cancellations are concentrated, tests three possible explanations, and proposes one targeted intervention with a quantified impact estimate.

A1: Cancellation Landscape

Cancellation rates were analyzed across two dimensions at once: **property city and check-in month**. The heatmap below shows cancellation rates for every city-month combination. Darker cells indicate higher rates.



Goa during the monsoon months (June to August) is the clearest hotspot. Several other city-month combinations also show elevated rates, but Goa's monsoon period stands out as the single most visible concentration.

The main takeaway is that cancellations are not evenly spread. They cluster around specific cities and months, which makes targeted intervention more practical than a platform-wide policy change.

A2: Rate vs Volume Analysis

A segment with a high cancellation rate is not necessarily the same as a segment with a high cancellation count. Both need to be examined separately to understand what will actually move the platform-wide number.

Segments were defined as city and booking channel combinations with at least 30 bookings. Each was ranked by cancellation rate and by absolute cancellation count.

Top 3 by rate: OTA bookings in Goa and similar high-risk city-channel combinations have the highest cancellation rates per booking.

Top 3 by volume: OTA bookings in Bangalore, Chennai, and Pune generate the largest absolute number of cancellations. Their rates are elevated but not as extreme, yet the booking volumes are much larger.

The two lists do not fully overlap. This distinction matters for the board's 5 percentage point target. Fixing only the highest-rate segments may have limited impact if those segments are small. High-volume segments will move the platform-wide number more even with a moderate rate reduction.

A3: Root Cause Analysis

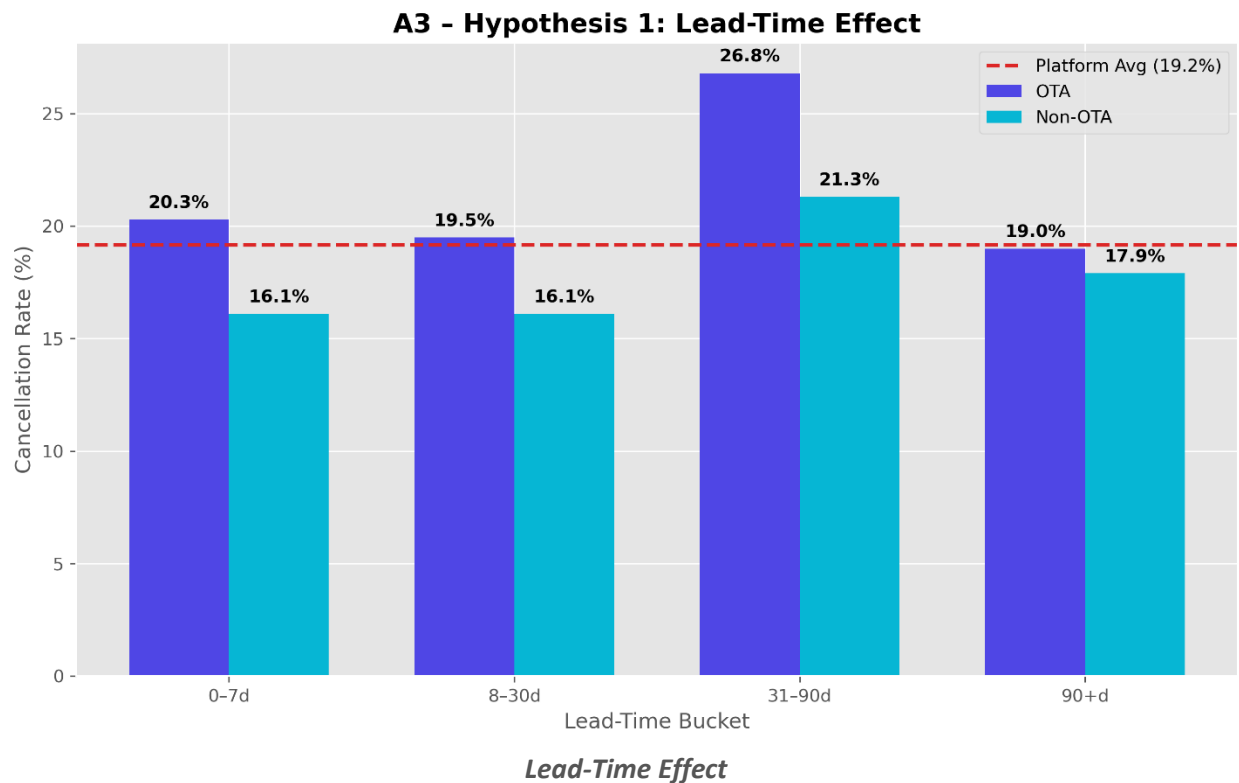
A2 showed that OTA bookings appear in both the highest-rate and highest-volume segments. Three hypotheses were tested to identify the underlying cause.

Baseline: OTA bookings cancel at a higher rate than Non-OTA bookings across the platform.

H1: Lead-Time Effect

Hypothesis: OTA's higher cancellation rate is caused by customers booking further in advance.

OTA and Non-OTA cancellation rates were compared across four lead-time buckets: 0-7 days, 8-30 days, 31-90 days, and 90+ days. The median lead time is 15 days for both groups.



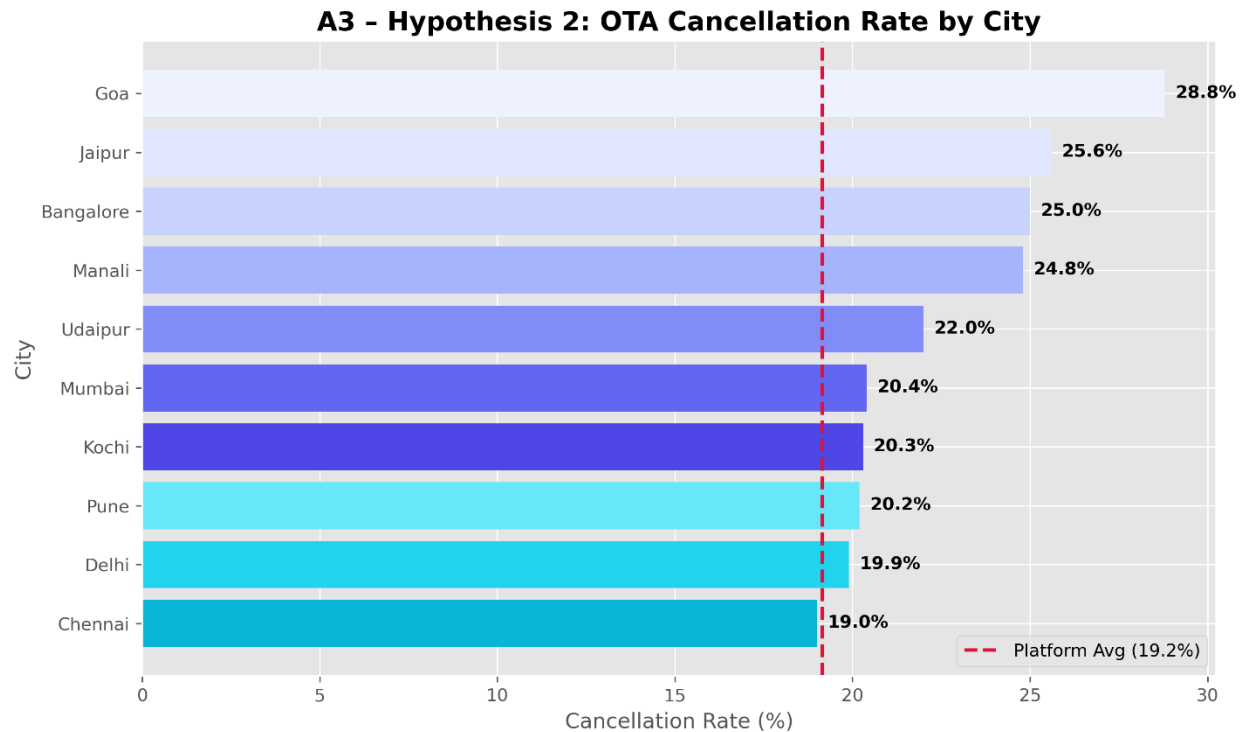
OTA rates are higher than Non-OTA in every lead-time bucket. The largest gap appears in the 31-90 day bucket. However, since the gap appears across all buckets and lead times are identical on median, lead time alone does not explain why OTA cancels more.

Verdict: Partially supported. Lead time is useful for targeting the highest-risk sub-segment but is not the root cause.

H2: Channel Mix Effect

Hypothesis: OTA's rate is being pulled upward by one or two city outliers.

OTA cancellation rates were compared across all cities. If this hypothesis were correct, removing a couple of cities would explain the gap.



OTA cancellation Rate by City

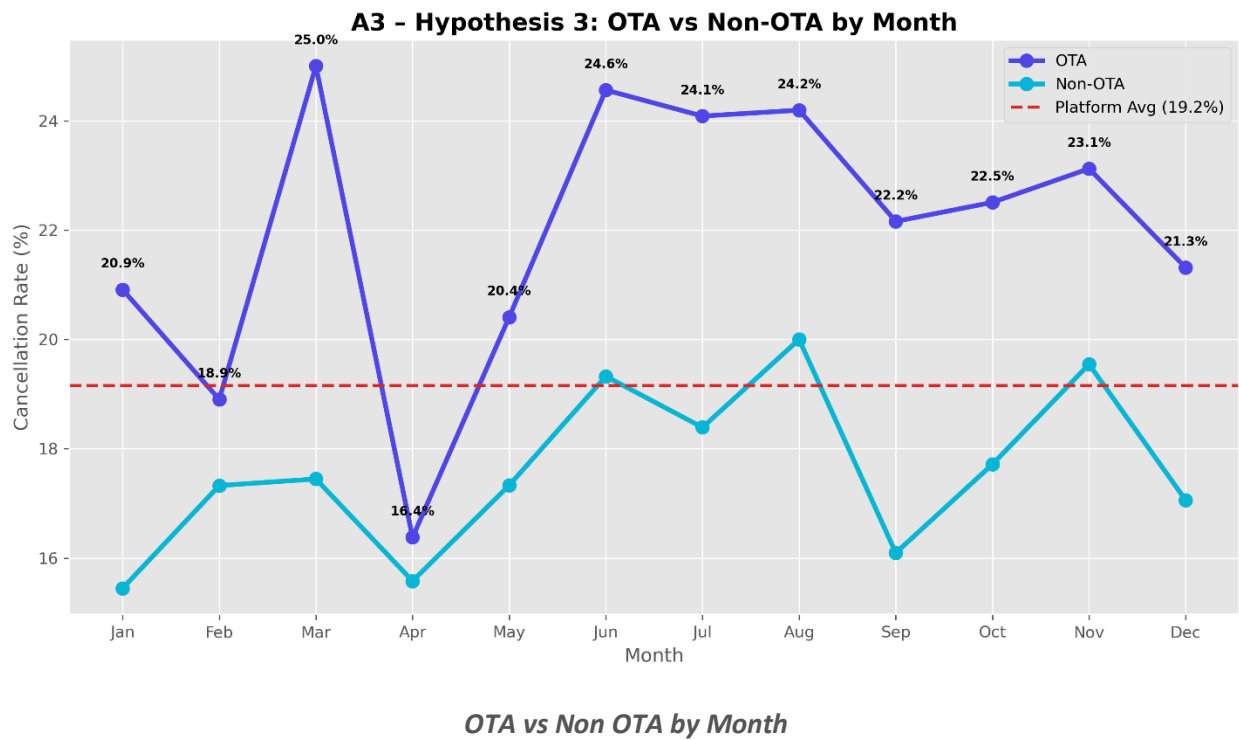
OTA rates are elevated across most cities. Goa, Bangalore, and Jaipur are particularly high, but the pattern is not limited to them. OTA exceeds the platform average in multiple locations.

Verdict: Not supported. The issue is not geographic.

H3: Structural OTA Effect

Hypothesis: OTA's higher rate is driven by specific months or seasons.

OTA and Non-OTA cancellation rates were compared month by month across the year.



OTA cancellation rates stay above Non-OTA rates throughout most months. The gap does not disappear in any particular season.

Verdict: Not supported. The gap is not seasonal.

Final Verdict

Hypothesis	Verdict
H1: Lead-Time Effect	Partial. Useful for targeting but not the root cause
H2: Channel Mix Effect	Ruled out. Not driven by city outliers
H3: City-and-Season Effect	Ruled out. Gap persists throughout the year
Structural OTA Effect	Best supported. OTA cancellation rate consistently exceeds Direct Website across cities and months

OTA is the strongest driver of elevated cancellations. The 31-90 day lead-time bucket is the highest-priority sub-segment because it combines a high cancellation rate with a meaningful cancellation count.

A4: Recommendation

WHO

All OTA bookings platform-wide, with priority on the 31-90 day lead-time segment across the five highest-volume cities: Bangalore, Chennai, Pune, Udaipur, and Mumbai.

WHAT

Introduce a tiered non-refundable deposit for OTA bookings. Customers booking 31-90 days before check-in pay a 20-25% non-refundable deposit at the time of booking. Shorter lead-time bookings stay flexible.

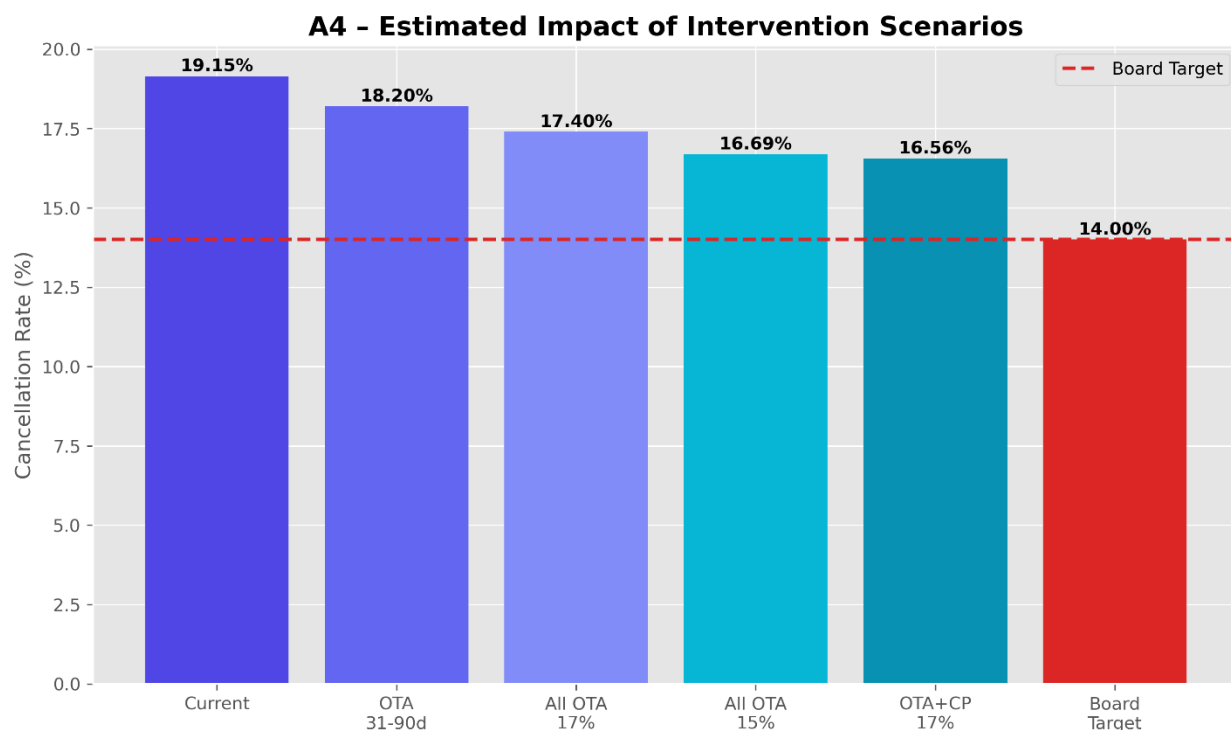
WHY

The OTA cancellation rate is 22.0% versus 14.8% for Direct Website bookings. This gap holds across cities and months, which points to a structural channel effect rather than a city-specific or seasonal issue. The 31-90 day OTA segment contains 341 cancellations at a 26.8% rate, making it the most efficient segment to target.

Counterfactual

If no action is taken, the platform cancellation rate stays near 19.15%. The strongest OTA-only intervention reduces this to approximately 16.69%. The combined OTA and Corporate Portal intervention brings it to 16.56%. Neither scenario fully reaches the board's 14% target, so additional measures would be needed to close the remaining gap.

Estimated Impact



Estimated reduction: -1.75 to -2.46 percentage points platform-wide. This would bring the overall cancellation rate from approximately 19.15% down to between 16.7% and 17.4%, preventing approximately 207-291 cancellations.

Risk

OTA partners may resist deposit requirements, or customers may shift to competitors with more flexible policies. A pilot rollout in a small number of high-volume cities is recommended before going platform-wide

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Weather Cancellation Analyzer

This mini-project extends the hotel bookings dataset by pulling in real historical weather data and checking whether weather conditions on the check-in date are associated with higher cancellation or no-show rates.

The analysis focuses on Goa and Manali bookings from 2024, since these two cities have very different weather patterns and together give a reasonable sample to work with. After cleaning, the working dataset had 1,582 bookings: 876 from Goa and 706 from Manali.

Weather data was fetched from the Open-Meteo Historical Weather API, which is free and requires no API key. Instead of calling the API once per booking row, I grouped bookings by city and date first, which reduced the number of API calls to 649 unique city-date combinations. The API returned daily precipitation totals and maximum temperature for each combination. All 649 requests succeeded with no missing values.

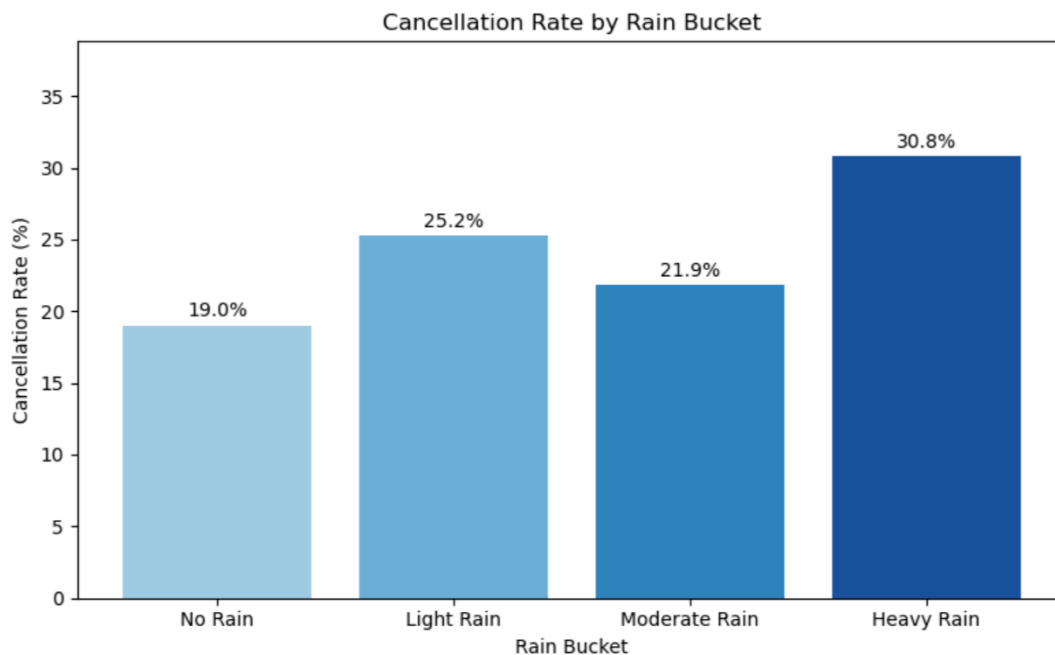
The weather data was then merged back onto the booking records using city and check-in date as join keys. The merge preserved all 1,582 rows without any duplication or data loss.

How the Analysis Works

Rainfall was grouped into four buckets: No Rain (0 mm), Light Rain (up to 2.5 mm), Moderate Rain (up to 10 mm), and Heavy Rain (above 10 mm). Temperature was similarly grouped into Cool (below 20°C), Normal (20-30°C), Hot (30-35°C), and Extreme (above 35°C). Cancellation and no-show rates were then calculated separately for each bucket.

Key Insights

Rainfall is more strongly associated with cancellations than temperature.



The cancellation rate varied by 11.78 percentage points across rainfall categories, compared to only 6.63 percentage points across temperature categories. This suggests that rain is the more useful signal when thinking about weather-related cancellation risk.

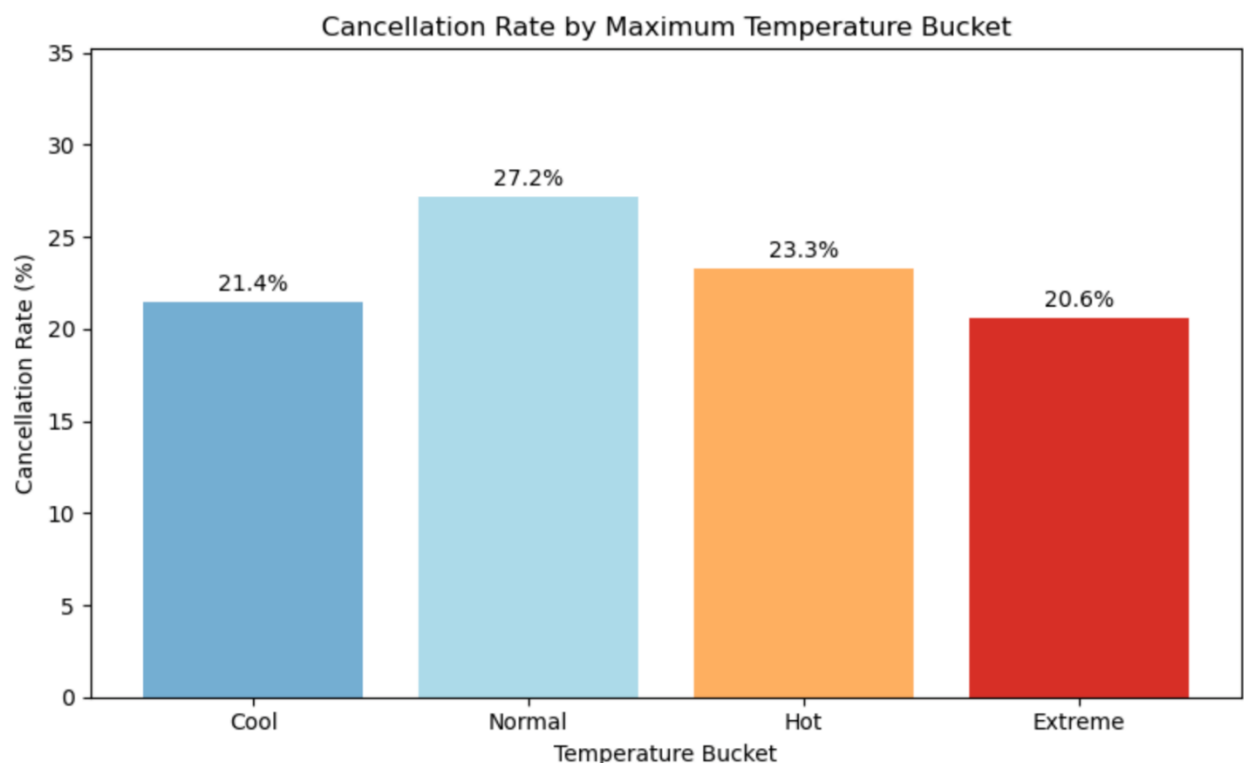
Heavy rain days show a noticeably higher cancellation rate.

On heavy-rain days (above 10 mm), the cancellation rate was 30.81%, compared to 21.38% on all other days. That is a difference of 9.43 percentage points. This gap cannot be spotted from the booking data alone, which is what makes the weather enrichment useful.

The effect is concentrated in Goa, not Manali.

When the heavy-rain comparison is split by city, Goa shows a large jump: 37.50% cancellation rate on heavy-rain days versus 21.70% on other days, a gap of 15.8 percentage points. In Manali, the difference is much smaller: 21.98% versus 20.99%, practically no change. This makes sense given that Goa is a beach destination where rain directly disrupts the purpose of the trip, while Manali attracts visitors who are often expecting cold or snowy conditions and are less deterred by precipitation.

Cancellation rates across the four temperature buckets ranged from 20.58% (Extreme heat) to 27.21% (Normal temperature). The fact that Normal temperature has the highest cancellation rate in this group suggests that temperature is not a reliable standalone predictor of cancellations in this dataset.



What This Adds to the Analysis

The dataset alone could tell us that Goa has a higher cancellation rate during monsoon months (June to August), which was already identified in Section 2. What the weather data adds is a more granular explanation: it is specifically heavy-rain days within those months that drive the spike, not just the season label itself.

This means a platform could theoretically flag bookings for Goa when the forecast shows heavy rain around the check-in date and proactively offer customers a free date-change. That kind of intervention is only possible if you have the weather data linked at the booking level, which is what this project demonstrates.

Design Decisions and Limitations

API requests were batched by city and date range rather than sent one per booking row, which kept the total number of calls manageable and avoids rate-limiting issues. Each city was mapped to a single coordinate, which is a simplification since properties within a city can be spread across different microclimates.

The main limitation is that this analysis shows association, not causation. It is possible that certain types of customers or certain booking channels are more likely to book on rainy days in Goa, and that those underlying factors are what actually drive the higher cancellation rate rather than the rain itself. A proper causal analysis would need to control for those variables. For now, the finding is best treated as a signal worth investigating further, not a definitive conclusion.